

RETAIL SALES FORECASTING & CUSTOMER PURCHASE ANALYSIS

Data Science Internship – Minor Project

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Project	Retail Sales Forecasting & Customer Purchase Analysis
Tools	Python, Pandas, NumPy, Matplotlib, Scikit-learn, Power BI
Submission Deadline	August 17, 2026

This report presents the complete data science workflow for a retail sales forecasting project, from data preprocessing and exploratory analysis to machine learning, forecasting, business recommendations, and Power BI reporting.

1. Executive Summary

The Retail Sales Forecasting & Customer Purchase Analysis project was developed to help a retail business understand historical sales performance, customer purchasing behavior, product and store performance, and expected future sales. The project follows a complete data science workflow: data collection, preprocessing, exploratory data analysis, feature engineering, predictive modeling, forecasting, visualization, and business reporting.

The project uses a retail transaction dataset containing transaction, date, store, customer, product, quantity, price, discount, sales, and profit information. A Random Forest Regressor was selected for monthly sales forecasting. The final results are presented through an interactive Power BI dashboard containing an executive overview, sales forecast, customer analysis, and business insights.

2. Business Problem

The retail chain experiences fluctuations in sales across products and store locations. Management needs a data-driven method to identify high-performing categories and locations, understand customer purchasing behavior, recognize seasonal patterns, and estimate future sales.

Project Objectives

- Clean and prepare historical retail transaction data.
- Analyze monthly sales, product categories, store locations, and customer behavior.
- Identify relationships among sales, quantity, price, discounts, and profit.
- Build and evaluate a machine learning model for sales forecasting.
- Generate sales predictions for the next three months.
- Develop a Power BI dashboard for management reporting.
- Provide recommendations for inventory, promotions, staffing, and planning.

3. Dataset Description

The project uses a retail sales transaction dataset. The available fields include Transaction_ID, Date, Store_ID, Store_Location, Customer_ID, Customer_Age, Customer_Gender, Product_ID, Product_Category, Product_Name, Quantity, Unit_Price, Discount_Percent, Payment_Method, Sales_Amount, and Profit.

Field	Purpose
Transaction_ID	Unique transaction identifier
Date	Transaction date used for time-based analysis
Store_Location	Store-level performance analysis
Customer_Age / Gender	Customer behavior analysis
Product_Category / Product_Name	Product performance analysis
Quantity	Units sold
Unit_Price	Price per unit
Discount_Percent	Discount applied to the transaction
Sales_Amount	Original transaction sales amount
Profit	Transaction profit
Payment_Method	Customer payment preference

Data availability note: The dataset does not contain a Customer_Segment field. Customer behavior was therefore analyzed using available attributes such as age, gender, and payment method rather than inventing a segment field.

4. Milestone 1 - Data Collection & Preprocessing

The dataset was imported using Pandas and inspected for structure and data quality. Missing values were checked and handled, duplicate records were removed, and the Date field was converted to datetime format.

Preprocessing Steps

- Imported the CSV dataset using Pandas.
- Inspected shape, columns, data types, and descriptive statistics.
- Checked and handled missing values.
- Removed duplicate transaction records.
- Converted Date to datetime format.
- Created Month, Quarter, Weekday, and Year features.
- Calculated Total Revenue after applying the percentage discount.
- Saved the cleaned dataset as retail_sales_cleaned.csv.

Revenue Calculation

Total Revenue was calculated as $\text{Sales Amount} \times (1 - \text{Discount Percent} / 100)$, providing a transaction-level revenue value after the recorded discount.

5. Milestone 2 - Exploratory Data Analysis

EDA was used to identify sales patterns, category performance, store performance, customer behavior, and relationships among numerical variables.

Monthly Sales Trend: Shows changes in revenue over time and helps identify growth, declines, and possible seasonality.

Product Category Performance: Compares total revenue across categories to identify strong and weak performers.

Revenue by Store Location: Compares store locations and highlights different levels of performance.

Customer Analysis: Uses customer gender, age, and payment method because Customer_Segment is not available.

Correlation Analysis: Examines relationships among quantity, unit price, discount, sales amount, profit, and revenue.

Major EDA Findings

- Monthly revenue varies over time, making historical trend analysis useful for planning.
- Product categories contribute differently to total revenue.
- Store locations show different revenue performance and may require location-specific strategies.
- Customer characteristics and payment methods provide useful information for targeted promotions.
- Sales Amount and Total Revenue are strongly related because Total Revenue is derived from Sales Amount and discount.

6. Milestone 3 - Predictive Modeling

Transaction data was aggregated into monthly sales. Time-based, lag, rolling-average, and seasonal features were created to capture recent sales behavior and recurring patterns.

Random Forest Regressor

- The Random Forest model combines multiple decision trees and can capture nonlinear relationships.
- Features include time index, month number, year, lagged sales, rolling average, and seasonal sine/cosine features.
- A chronological train/test split was used to avoid future-data leakage.
- Model performance was evaluated using MAE, MSE, RMSE, and R² Score.

Metric	Result
MAE	[Insert Python result]
MSE	[Insert Python result]
RMSE	[Insert Python result]
R ² Score	[Insert Python result]

Three-Month Sales Forecast

The trained Random Forest model was used to generate sales estimates for the next three months. The forecast output was saved as forecast_results.csv and imported into Power BI using Forecast_Month and Forecast_Sales.

Forecast Month	Forecast Sales
Month 1	[Insert forecast value]
Month 2	[Insert forecast value]
Month 3	[Insert forecast value]

Replace bracketed values with the exact values from the completed Python notebook before final submission.

7. Milestone 4 – Business Insights

Best-Selling Categories: Prioritize high-revenue categories for stock availability and monitor weaker categories for assortment or pricing changes.

Seasonal Trends: Use monthly historical trends and the three-month forecast to prepare for expected changes in demand.

Inventory Planning: Use historical demand and the forecast as a short-term guide for stock allocation and replenishment.

Promotional Opportunities: Target promotions using product performance, customer behavior, payment preferences, and demand patterns.

Store Strategy: Compare store revenue and investigate lower-performing locations to identify differences in demand or product mix.

Staffing: Align staffing levels with expected demand periods to improve service while controlling operating costs.

Business Recommendations

- Maintain sufficient inventory for high-performing categories.
- Use forecasts to support short-term purchasing and replenishment.
- Investigate low-performing stores and categories instead of relying only on overall averages.
- Use customer purchasing patterns to create targeted promotions.
- Monitor discounts together with profit to protect profitability.
- Retrain and update the forecasting model as new sales data becomes available.

8. Milestone 5 – Power BI Dashboard

An interactive Power BI dashboard was designed to communicate the project results to business users. The dashboard is organized into three pages.

Page 1 – Executive Sales Overview

- Total Revenue KPI
- Total Profit KPI
- Units Sold KPI
- Total Transactions KPI
- Monthly Sales Overview
- Revenue by Product Category
- Store-wise Performance
- Year, Category, Store, and Payment Method slicers

Page 2 – Sales Forecast & Model Performance

- Total three-month forecast
- Average monthly forecast
- Highest and lowest forecast
- Sales Forecast – Next 3 Months chart
- Three-month forecast details table
- MAE, MSE, RMSE, and R^2 cards
- Forecast interpretation

Page 3 – Customer & Business Insights

- Customer Gender Distribution
- Payment Method Distribution
- Customer Age Distribution
- Top 10 Products by Revenue
- Key Business Insights
- Business Recommendations

9. Power BI Measures

The dashboard uses measures for the main KPIs. The revenue measure accounts for the percentage discount in the dataset.

Measure	Purpose
Total Revenue	Revenue after applying Discount_Percent
Total Sales	Sum of Sales_Amount
Total Profit	Sum of Profit
Total Quantity	Sum of Quantity
Total Transactions	Distinct count of Transaction_ID
Average Order Value	Total Revenue ÷ Total Transactions
Total Forecast Sales	Sum of Forecast_Sales
Average Forecast Sales	Average of Forecast_Sales

10. Conclusion

The project demonstrates an end-to-end data science workflow for retail analytics. Historical transaction data was cleaned and transformed, exploratory analysis was performed to understand sales and customer patterns, and a Random Forest Regressor was developed for monthly sales forecasting.

The three-month forecast provides a baseline for short-term inventory, staffing, and promotional planning. The Power BI dashboard transforms the technical results into a management-friendly report covering KPIs, category and store performance, forecast results, customer analysis, and business recommendations.

Future improvements could include additional historical data, holiday and promotion variables, alternative forecasting models, hyperparameter tuning, and automated model retraining.

11. Final Deliverables Checklist

Deliverable	File
Jupyter Notebook	retail_sales_forecasting.ipynb
Cleaned Dataset	retail_sales_cleaned.csv
Forecast Results	forecast_results.csv
Power BI Dashboard	Retail_Sales_Forecasting_Dashboard.pbix
Dashboard Screenshots	dashboard_page1.png, dashboard_page2.png, dashboard_page3.png
Final Report	Retail_Sales_Forecasting_Final_Report.pdf

